POWER BI PROJECT DOCUMENTATION

Electronics Retailer Analysis

Sales & Performance Analytics Dashboard

PREPARED BY	**Wilfred Chizurukeme P. / Maybach Group member**
TOOL	Microsoft Power BI Desktop (DAX and Power Query)
DATA SOURCE	GIZ ICT Tech Training: Customers, Sales, Stores, Products and Exchange Rates
ANALYSIS PERIOD	2016 to 2021 (orders from 1 January 2016 to 20 February 2021)
DATE	November 2026

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1. Project Overview

The Electronics Retailer Analysis is a Power BI sales and performance analytics project built by Wilfred Chizurukeme, a member of Maybach Group. It brings sales, product, customer, store and delivery data for an electronics retail business into one interactive dashboard, so that commercial and finance stakeholders can monitor revenue, profitability and operational performance from a single place.

The dashboard moves from a company-wide summary to the questions that drive decisions: which brands and products earn the most, which store locations and sales channels perform best, who the customers are, how quickly online orders are delivered, and how performance has changed month by month and year by year.

Project at a glance

Item	Detail
Project title	Electronics Retailer Analysis
Team	Maybach Group
Prepared by	Wilfred Chizurukeme P. / Maybach Group member
Tool	Microsoft Power BI Desktop
Data source	Five CSV files supplied through the GIZ ICT tech training
Analysis period	2016 to 2021 (order dates run from 1 January 2016 to 20 February 2021)
Headline results	Total sales $55,755,480; product cost $23,092,791; gross profit $32,662,688 (58.6% margin); 26,326 orders; 197,757 units sold
Dashboard	Interactive report with six pages (Home, Sales Overview, Sales Analysis, Products Summary, Customer Demographics and Logistics) and two tooltip pages

2. Problem Statement

Retail and commercial teams typically track revenue, cost and customer data across disconnected spreadsheets and point reports. This makes it hard to see a consistent, current picture of business performance.

There is a need for one interactive tool that lets stakeholders filter by year, month, sales channel and country, and immediately see the effect on revenue, profit, order volume, customer behaviour and delivery performance, without waiting for a manual report cycle.

> **How this project responds**
> A self-service Power BI dashboard that lets users move from a company-wide summary to brand-, product-, store- and customer-level detail, with a dedicated logistics view for online orders.

3. Aim and Objectives

> **Aim**
> To conduct a comprehensive analysis of sales transaction data to evaluate business performance, identify the key drivers of revenue and profitability, and generate data-driven insights that support strategic decision-making.

Specific objectives

- Examine historical sales and revenue trends and identify significant changes in business performance.
- Evaluate profitability and compare profit with revenue across different business dimensions.
- Identify high-performing and underperforming brands and products based on revenue and profit.
- Assess sales and profit distribution across countries, states and physical store locations.
- Compare the performance of different sales channels, particularly online and physical retail.
- Analyse customer demographic characteristics and their relationship with sales performance.
- Evaluate month-over-month and year-over-year changes in sales and profitability.
- Monitor delivery lead times for online orders.
- Develop an interactive Power BI dashboard for monitoring and communicating key performance indicators.
- Translate analytical findings into actionable insights and recommendations.

4. Data Sources and Data Model

The facilitator of the GIZ ICT tech training provided five key CSV files: Customers, Sales, Stores, Products and Exchange Rates. They were loaded and cleaned in Power Query, then modelled as a star schema around one fact table (Sales). The model also holds a calculated date table, a currency table, a dedicated measures table and a calculated product-and-store summary table. Figure 1 shows the tables and how they are related.

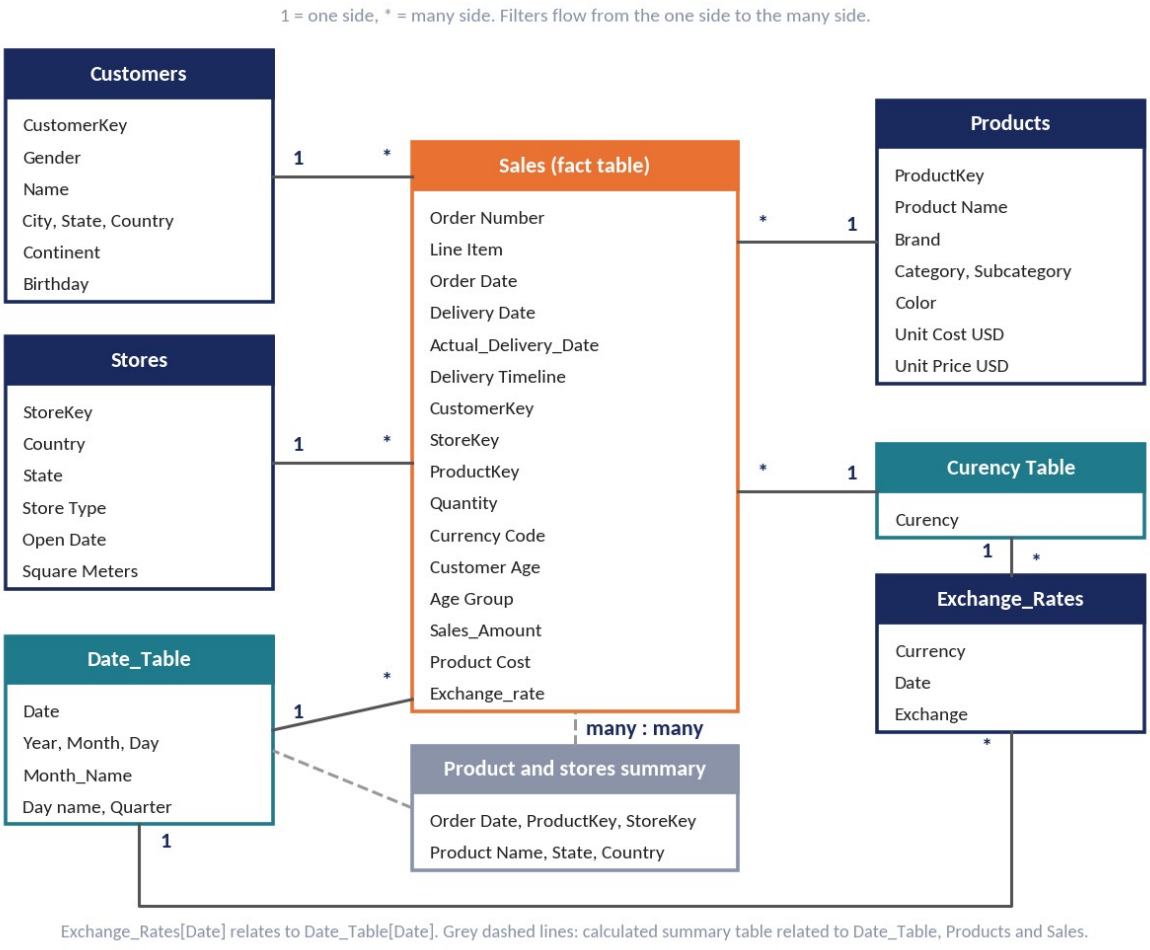


Figure 1: *Data model of the Electronics Retailer report, showing the Sales fact table and its related tables*

Table	Role	Key fields
Sales	Fact table (62,884 rows, 26,326 orders)	Order Number, Line Item, Order Date, Delivery Date, Actual Delivery Date, Quantity, Currency Code, Delivery Timeline; calculated columns Sales_Amount, Product Cost, Exchange_rate, Customer Age, Customer Age (bins) and Age Group
Customers	Dimension (15,266 customers)	CustomerKey, Gender, Name, City, State, Country, Continent, Birthday
Products	Dimension (2,517 products)	ProductKey, Product Name, Brand, Category, Subcategory, Color, Unit Cost USD, Unit Price USD
Stores	Dimension (67 stores: 66 physical and 1 online)	StoreKey, Country, State, Store Type (calculated), Open Date, Square Meters
Date_Table	Calculated date dimension (1 January 2016 to 20 February 2021)	Date, Year, Quarter, Month, Month name, Day, Day name. Month name is sorted by the Month number
Exchange_Rates and Curency Table	Currency conversion	Currency, Date, Exchange; list of currencies from Sales and Exchange Rates
Product and stores summary table	Calculated summary table	Order Date, ProductKey, Product Name, StoreKey, State, Country
Measure	Measures table	All DAX measures used by the report pages

The Sales table is related to Customers (CustomerKey), Products (ProductKey), Stores (StoreKey) and Date_Table (Order Date) through one-to-many relationships with single-direction filtering. Sales_Amount and Product Cost are calculated from the product list price and unit cost in US dollars, so every value in the report is in USD. The report pages read from the Sales, Customers, Products, Stores and Date_Table tables; the exchange-rate and summary tables support currency conversion and further analysis and are not used by any page.

Keeping measures in a dedicated table separates calculations from the raw fact table, keeps the model tidy, and makes each measure easy to find and reuse across every report page.

5. Tools and Technologies

Tool	Purpose
Power BI Desktop	Data modelling, DAX measures, report and dashboard design
Power Query	Loading, cleaning and shaping the five CSV source files
DAX	Measures, calculated columns and calculated tables
Star schema modelling	One fact table related to dimension tables for clear, fast filtering
Slicers, buttons and page navigation	Year, month, sales channel and country filters; Clear all Filters buttons; a page navigator on every page
Tooltip pages	Report-page tooltips that show extra detail when a user hovers over a chart

6. Formulas and Functions (DAX)

The dashboard is powered by the following DAX logic, shown exactly as it is written in the Power BI file. Measure names are given as they appear in the model.

Core measures

Measure	DAX formula
Sales	Sales = SUM('Sales'[Sales_Amount])
Expensese (shown as Product Cost on the dashboard)	Expensese = SUM(Sales[Product Cost])
Profit	Profit = [Sales] - [Expensese]
Profit Margin	Profit Margin = DIVIDE([Profit], [Sales], 0)
Orders	Orders = DISTINCTCOUNT(Sales[Order Number])
Quantity Sold	Quantity Sold = SUM(Sales[Quantity])
Average Product Price	Average Product Price = AVERAGE(Products[Unit Price USD])
Sales % of Total	Sales % of Total = DIVIDE([Sales], CALCULATE([Sales], ALL('Stores'[Store Type])))
Onlines sales	CALCULATE([Sales], Stores[Store Type] = "Online")
Physical store sales	CALCULATE([Sales], Stores[Store Type] = "Physical")

Sales % of Total removes the channel filter in the denominator, so each channel is always shown as a share of all-channel sales, even when the report is filtered to one channel.

Time intelligence

Measure	DAX formula
Previous Month Sales	CALCULATE([Sales], DATEADD(Date_Table[Date], -1, MONTH))
MoM Sales Change	[Sales] - [Previous Month Sales]
MoM Sales %	Pools every month in view with its previous month and compares the totals (full formula below)
MoM sales Display	Shows the MoM Sales % with an up or down arrow and a +/- sign, or a dash when blank (full formula below)
LM Profit	CALCULATE([Profit], PREVIOUSMONTH(Date_Table[Date])) + 0
Profit Variance M	[Profit] - [LM Profit]
MoM % Profit Growth	DIVIDE([Profit Variance M], [LM Profit], 0)
LY sales	CALCULATE([Sales], PREVIOUSYEAR('Date_Table'[Date])) + 0
Year Varaince	[Sales] - [LY sales]
YoY % Growth	DIVIDE([Year Varaince], [LY sales], 0)

Measure	DAX formula
LY Profit	CALCULATE([Profit], PREVIOUSYEAR('Date_Table'[Date])) + 0
Profit Variance	[Profit] - [LY Profit]
YoY % Profit Growth	DIVIDE([Profit Variance], [LY Profit], 0)

MoM Sales % is written so that it works with a single month, a selected year or no selection at all. It lists every year-and-month in view, looks up the sales of the month before each one, drops months that have no previous month, and compares the totals. In the month-over-month table, each column (January to December) therefore compares all Januaries with the Decembers before them, all Februaries with the Januaries before them, and so on.

```
MoM Sales % =
VAR Months =
    ADDCOLUMNS (
        SUMMARIZE ( Date_Table, Date_Table[Year], Date_Table[Month] ),
        "@Idx", Date_Table[Year] * 12 + Date_Table[Month],
        "@Cur", [Sales]
    )
VAR WithPrev =
    ADDCOLUMNS (
        Months,
        "@Prev",
            VAR i = [@Idx]
            RETURN
                CALCULATE (
                    [Sales],
                    REMOVEFILTERS ( Date_Table ),
                    FILTER (
                        ALL ( Date_Table[Year], Date_Table[Month] ),
                        Date_Table[Year] * 12 + Date_Table[Month] = i - 1
                    )
                )
    )
VAR Valid = FILTER ( WithPrev, NOT ISBLANK ( [@Prev] ) )
VAR CurTotal = SUMX ( Valid, [@Cur] )
VAR PrevTotal = SUMX ( Valid, [@Prev] )
RETURN
    DIVIDE ( CurTotal - PrevTotal, PrevTotal )

MoM sales Display =
VAR m = [MoM Sales %]
RETURN
    IF ( ISBLANK ( m ), "—",
        IF ( m >= 0, "▲ ", "▼ " ) & FORMAT ( m, "+0.0%;-0.0%" ) )
```

Logistics measures

Measure	DAX formula
Average Delivery Days	CALCULATE(AVERAGE(Sales[Delivery Timeline]), Sales[Delivery Timeline] > 0)
deliver when	IF(ISBLANK([Average Delivery Days]), BLANK(), FORMAT([Average Delivery Days], "0.0") & " Days")

Delivery Timeline is only recorded for online orders, so the filter greater than zero keeps the average to orders that were actually delivered. The deliver when measure turns the average into the "4.5 Days" label used on the Logistics page and its tooltip.

Calculated columns and tables

Item	Type	Logic
Sales_Amount	Calculated column	Sales[Quantity] * RELATED(Products[Unit Price USD])
Product Cost	Calculated column	Sales[Quantity] * RELATED(Products[Unit Cost USD])
Customer Age	Calculated column	DATEDIFF(RELATED(Customers[Birthday]), Sales[Order Date], YEAR): the customer's age at the order date
Customer Age (bins)	Calculated column	INT(Customer Age / 10) * 10, blank when age is blank
Age Group	Calculated column	SWITCH(TRUE(), ...) grouping ages into 10–19, 20–29, 30–39, 40–49, 50–59, 60–69, 70+ and Unknown
Store Type	Calculated column	IF(Stores[State] = "Online", "Online", "Physical")
Exchange_rate	Calculated column	Finds the latest exchange rate on or before the order date for the sale currency (VAR, FILTER, MAXX)
Date_Table	Calculated table	CALENDAR(MIN(Sales[Order Date]), MAX(Sales[Order Date])), with Year, Month, Day, Month_Name (MMM), day name and quarter columns
Curency Table	Calculated table	DISTINCT(UNION(...)) of the currencies in Sales and Exchange_Rates
Product and stores summary table	Calculated table	SUMMARIZE(Sales, Order Date, ProductKey, Product Name, StoreKey, State, Country)

7. Dashboard Walkthrough

The dashboard is organised into six pages that each answer a different stakeholder question, from "how are we doing overall?" to "which brands, customers and deliveries drive performance?". A page navigator at the top of every page links them together. Two hidden tooltip pages add detail when a user hovers over a chart.

7.1 Home

The landing page. A branded cover with a page navigator that lists all six pages, so the report can be navigated like an app. The current page is highlighted in the navigator.



Figure 2: Home page with the page navigator

7.2 Sales Overview

The headline page for company-wide performance. Seven KPI cards sit on the left, and the Year tiles and month drop-down filter the cards and charts. The Clear all Filter button resets every selection.

Visual	Type	Fields shown
KPI cards	Cards	Total Sales ($55,755,480), Product Cost ($23,092,791), Profit ($32,662,688), Profit Margin (59%), Total Orders (26,326), Quantity Sold (197,757) and Total Customers (15,266)
Sales Amount Trend	Line chart	Sales by year, 2016 to 2021
Year Over Year % Growth	Card	YoY % Growth for the selected year
Sales by Month	Area chart	Sales by month, January to December
Month-Over-Month Sales Growth	Table	Month name with MoM sales Display (green and red arrows)
Year tiles and month drop-down	Slicers	Year and Month_Name from the date table



Figure 3: Sales Overview page

7.3 Sales Analysis

Explains where revenue and profit come from: by sales channel, year, store location, brand and product. A strip of four cards tracks YoY and MoM growth, and four slicers (Year, Month, Channels and Store country) filter the page.

Visual	Type	Fields shown
Sales % of Total by Sales Channel	Donut chart	Store Type against Sales % of Total: Online 20.5% ($11,404,325) and Physical 79.5% ($44,351,155)
Growth strip	Cards	YoY % Profit Growth, YoY % Growth, MoM % Profit Growth and MoM Sales %
Sales and Profit by Year	Combo chart	Year with Sales (columns) and Profit (line); hovering shows the tooltip in Figure 5
Sales and Profit by Store Location (State)	Bar chart	State against Sales and Profit
Sales and Profit by Brand	Bar chart	Brand against Sales and Profit
Top 5 products by sales and profit	Bar chart	Product Name against Sales and Profit (top five by sales)

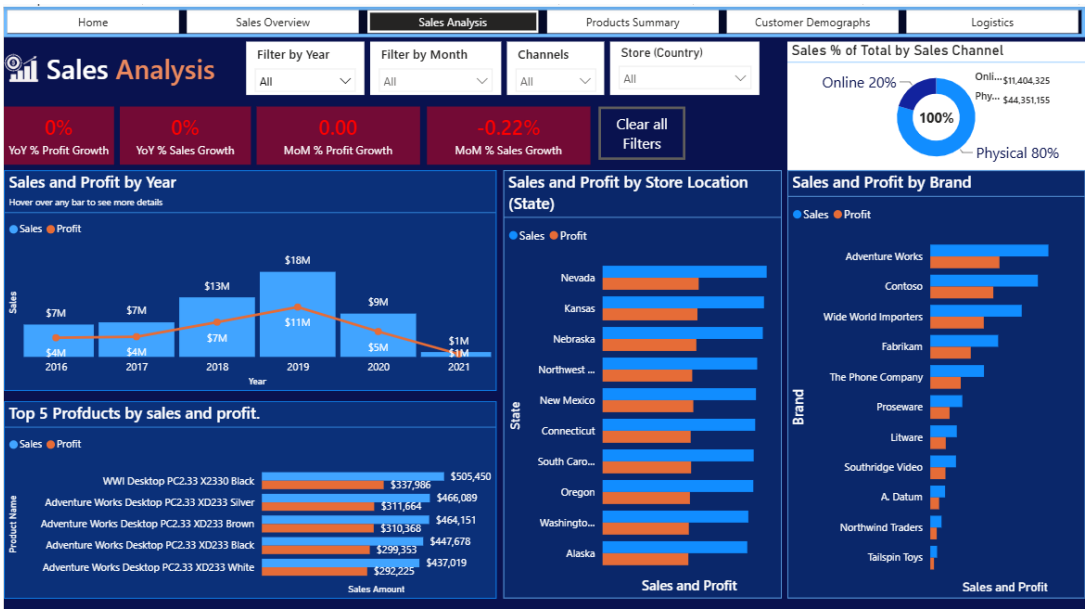


Figure 4: Sales Analysis page

Hovering over a year in the Sales and Profit by Year chart opens a tooltip page that shows Total Sales and Gross Profit next to a table of each month's sales change and MoM profit growth.

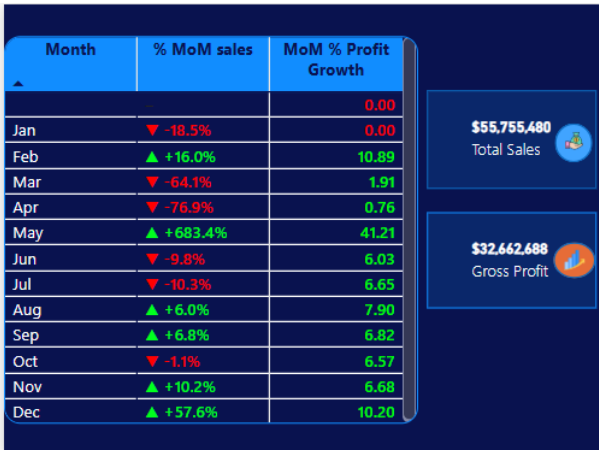


Figure 5: Tooltip shown when hovering over Sales and Profit by Year

7.4 Products Summary

Shows how each brand performs on price, volume, range and orders. A Year slicer, a Month_Name slicer and a Sales Channel filter control the page.

Visual	Type	Fields shown
Average Product Price by Brand	Column chart	Average of Unit Price USD by brand; highest Northwind Traders ($812), lowest Tailspin Toys ($48)
Quantity Sold by Brand	Column chart	Units sold by brand; Contoso leads with about 50K
Total Available Product by Brand	Column chart	Number of distinct products in the catalogue by brand (not stock on hand); Contoso has the widest range (710 products)
Orders by Brand	Column chart	Distinct orders by brand; Contoso leads with about 12.2K



Figure 6: Products Summary page

7.5 Customer Demographics

Profiles who is buying: gender and age group, and how each segment contributes to orders and revenue. Age group is based on the customer's age on the order date.

Visual	Type	Fields shown
Total Count of Customers by Gender	Bar chart	Customers by gender: 7,748 male and 7,518 female
% Orders by Gender	Donut chart	Share of orders: 50.73% male and 49.27% female
% Sale contribution by Gender	Donut chart	Share of sales: 50.8% male and 49.2% female
% Sales Contribution by Age Group	Column chart	Share of sales by age group; 70+ is highest at 21.03%
Orders by Age Group	Column chart	Orders by age group; 70+ is highest at 5,498 orders

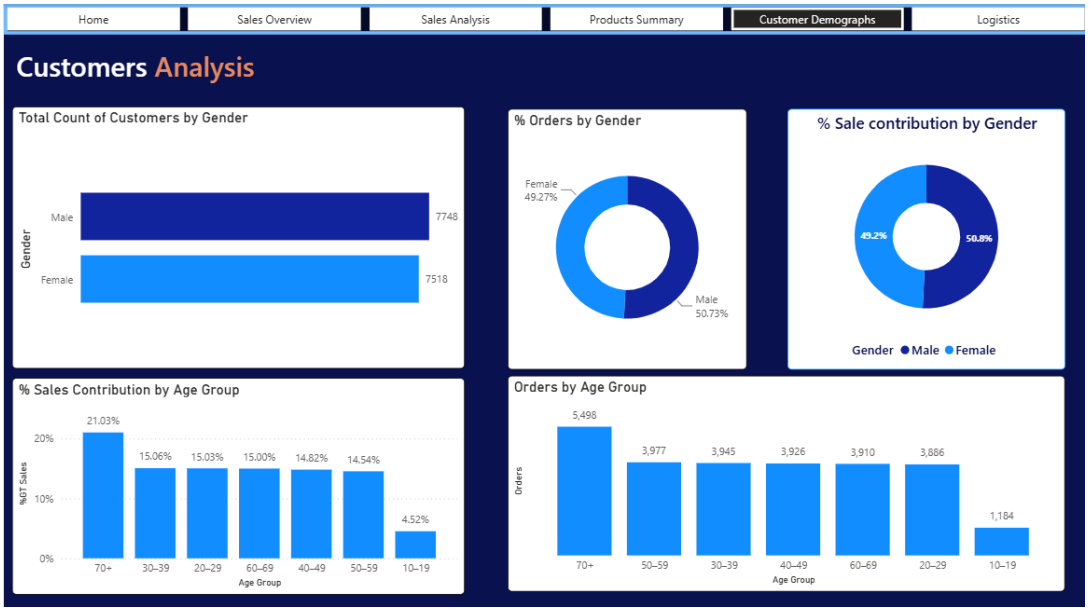


Figure 7: Customer Demographics page

7.6 Logistics

Looks at delivery performance. The page covers online orders only, because physical store purchases are collected at the point of sale and have no delivery time. A note on the page tells readers this, and a Year slicer filters the charts. The Sales table does not record where an order was placed or delivered, so lead time can currently be compared only by month, customer and product (see Recommendation 6).

Visual	Type	Fields shown
Lead Time card	Card	Average delivery days shown as text (4.5 Days) using the deliver when measure
Average Delivery Days by Month	Bar chart	Average delivery days for each month, January to December
Customer vs Lead Time	Column chart	Customer name against average delivery days; hovering shows the tooltip in Figure 9
Average Delivery Days of Product	Bar chart	Average delivery days by product name



Figure 8: Logistics page

Hovering over a bar in the Customer vs Lead Time chart opens a tooltip that shows the customer name, lead time, delivery point and order quantity.



Figure 9: Tooltip shown when hovering over Customer vs Lead Time

8. Key Insights

The figures below were calculated from the data in the Power BI model and match the values shown on the dashboard.

1. Revenue and profit performance

Sales grew every year from $6.95M in 2016 to a peak of $18.26M in 2019 (up 42.8% on 2018), then fell to $9.29M in 2020 (down 49.1%). Profit followed the same path, peaking at $10.70M in 2019.

The gross margin stayed between 58.4% and 59.1% in every year, so the fall after 2019 came from lower volume, not thinner margins: orders dropped from 9,083 in 2019 to 4,635 in 2020. The 2021 figure ($1.04M) covers only 1 January to 20 February and is not a full-year result.

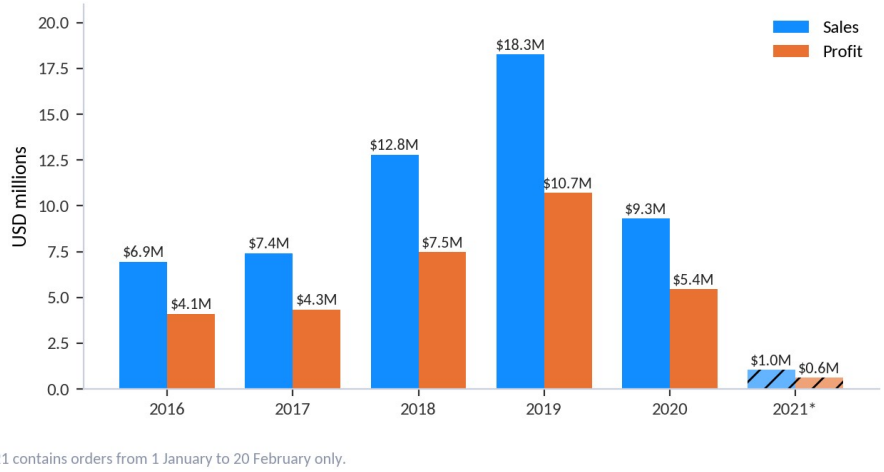


Figure 10: Sales and profit by year

2. Brand revenue

Adventure Works is the leading brand with $11.85M, or 21.3% of total revenue. Contoso follows with $10.79M (19.4%) and Wide World Importers with $9.17M (16.5%). Together the three brands contribute 57.1% of revenue.

Contoso sells the most units (49,827) and takes the most orders (12,155) at an average list price of $288. Northwind Traders has the highest average price ($812) but only 2.0% of revenue.

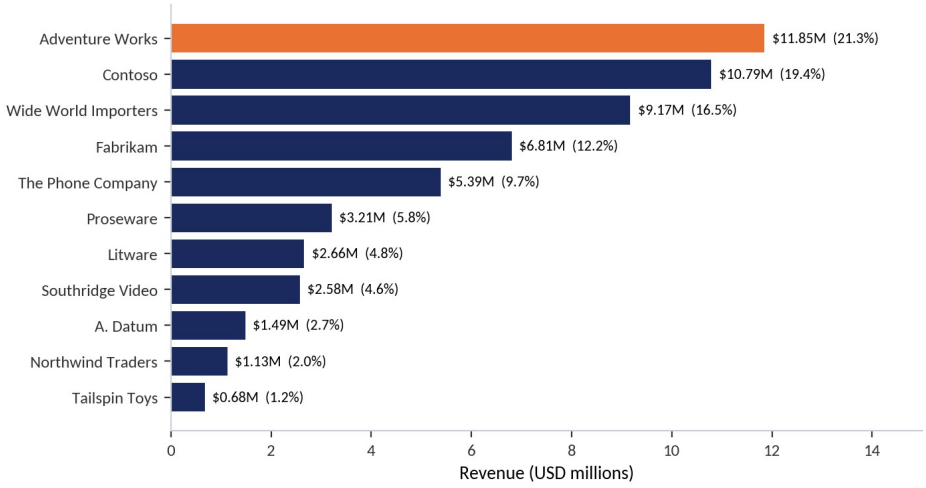


Figure 11: Revenue by brand and share of total revenue

3. Product revenue

Desktop PCs dominate the top five. The WWI Desktop PC2.33 X2330 Black earns the most, with $505,450 in revenue and $337,986 in profit. The next four are Adventure Works XD233 desktops: Silver ($466,089), Brown ($464,151), Black ($447,678) and White ($437,019).
Computers is the largest category with $19.30M, or 34.6% of revenue.

4. Revenue versus profitability

Higher revenue does not always mean a higher margin. Fabrikam earns a 59.7% margin on 12.2% of revenue, while The Phone Company earns 56.8% on 9.7%. Tailspin Toys has the lowest brand margin (54.6%) and Proseware the highest (60.2%), against a company average of 58.6%.

5. Sales channels

Physical stores generate $44.35M (79.5% of sales) and the online store $11.40M (20.5%). Margins are almost identical (58.6% physical and 58.5% online). The online share has grown steadily, from 16.8% in 2016 to 22.3% in 2020.

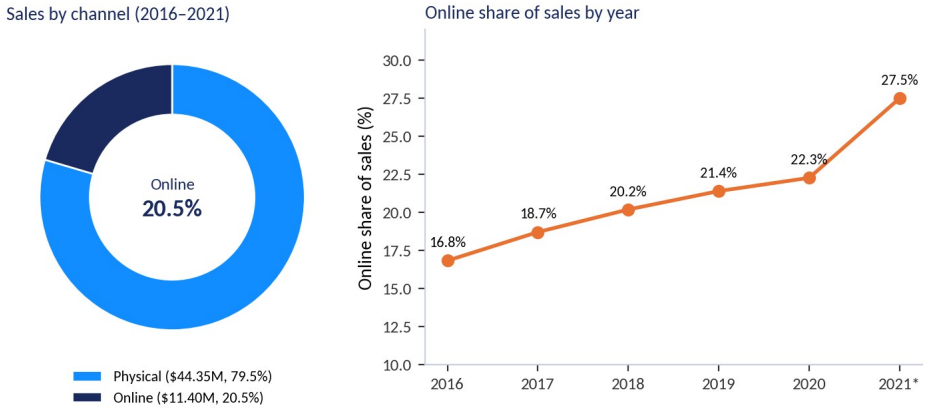


Figure 12: Sales by channel and online share of sales by year

6. Geographic profit contribution

By customer country, profit is concentrated in the United States at 53.55% of the total. The United Kingdom follows with 12.66%, then Germany with 9.75% and Canada with 8.44%.
Physical store sales are spread evenly across store states: Nevada ($1.42M), Kansas ($1.39M) and Nebraska ($1.38M) lead. The online store alone ($11.40M) is larger than any single state.

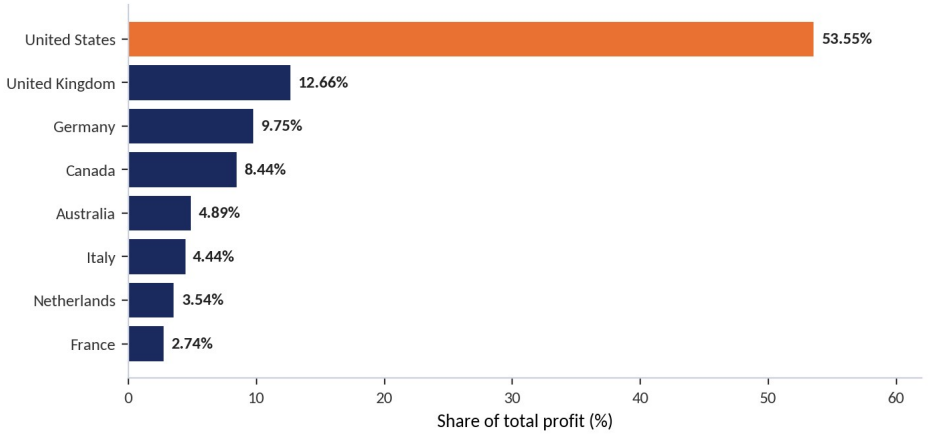


Figure 13: Share of total profit by customer country

7. Customers

The customer base is balanced: 50.75% of customers are male (7,748) and 49.25% female (7,518). Orders (50.73% and 49.27%) and sales (50.8% and 49.2%) split almost the same way, so there is no pronounced gender concentration.

Customers aged 70 and over are the largest segment, with 21.03% of sales and 5,498 orders. Each age group from 20 to 69 contributes between 14.5% and 15.1% of sales, and customers aged 10 to 19 contribute 4.52%.

8. Delivery lead time

Online orders take 4.5 days to deliver on average, and the average is steady across months (4.4 to 4.6 days). It has improved over time, from 7.3 days for 2016 orders to 4.0 days for 2020 orders. The slowest product is the Contoso Reserve Pen - Tablet Pen E200 Grey at 15 days.

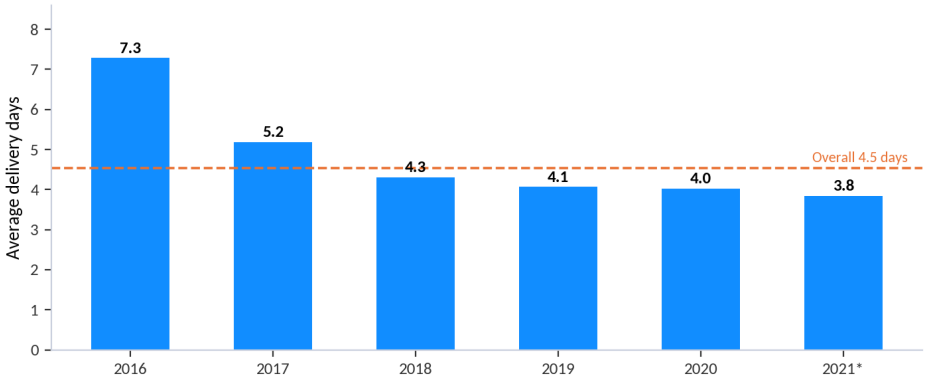
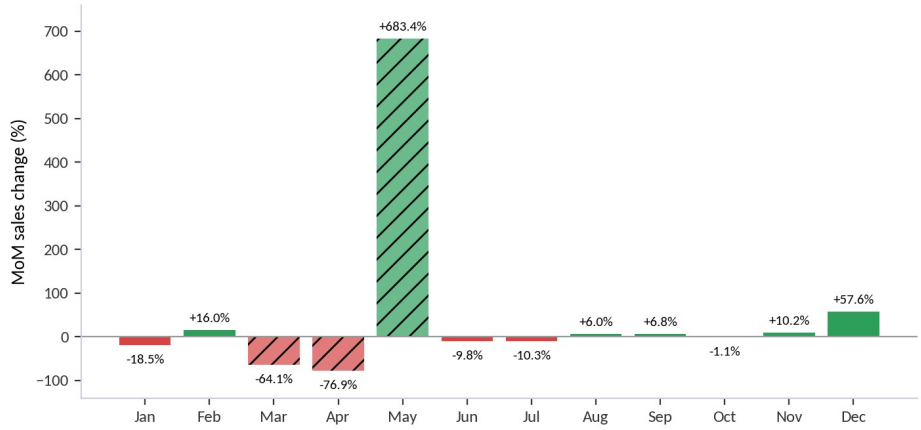


Figure 14: Average delivery days by order year (online orders)

9. Data Notes and Limitations

Readers should keep the following points in mind when using the dashboard.

- **Missing orders each spring.** No orders are recorded from about 17–19 March to 23–25 April in each year from 2016 to 2020 (35 to 37 days). This pulls March and April down and inflates May, so the month-over-month changes for March (-64.1%), April (-76.9%) and May (+683.4%) reflect missing data, not real performance (Figure 15).
- **2021 is a partial year.** Orders run only from 1 January to 20 February 2021 (46 days). The year-over-year fall for 2021 (-88.8%) compares two months with a full year and should not be read as a real decline.
- **Growth cards need a selection.** With no Year or Month selected, the YoY cards show 0% and the MoM cards show an all-period value. Select a year (and a month) to read a single-period change. The YoY measures use PREVIOUSYEAR, which returns the whole previous year, so they are most reliable on year-level views.
- **Profit is gross profit.** Sales are list price times quantity and cost is unit cost times quantity. Discounts, returns, tax and shipping are not in the data, so the 58.6% margin is a gross margin.
- **The online store is one record.** Online sales are held under a single store (State = Online), so "Online" can appear alongside states in store-location views.
- **Customer counts.** The Total Customers card counts every customer in the customer table (15,266). Of these, 11,887 placed at least one order and 4,547 bought online.
- **No inventory data.** The model has no stock table, so it cannot show how many units each store holds, when a product became available or left a store, or when to reorder. "Total Available Product by Brand" counts the products in the catalogue, not stock on hand, and products are linked to stores only through sales.
- **Customer age is approximate.** Age is the difference in calendar years between the birth year and the order year, so some customers may be a year older than their exact age.



Hatched bars (Mar–May) are distorted by the missing orders between about 17 March and 24 April in 2016–2020.

Figure 15: *Month-over-month change in sales by month name, pooled across 2016–2021*

10. Impact of the Dashboard

By consolidating sales, product, customer, store and delivery data into one interactive Power BI report, the dashboard replaces several disconnected spreadsheets with a single source of truth.

Stakeholders can filter by year, month, sales channel and store country and see the KPIs and charts respond at once, which supports faster decisions on pricing, stocking, store performance, customer targeting and delivery service.

11. Project Outcomes

1. **Centralised reporting.** Focused report pages replace manual, fragmented sales and finance reporting with one connected model.
2. **Faster decision-making.** Interactive slicers (year, month, sales channel, country) let stakeholders answer their own questions without waiting for a new report.
3. **Clear growth visibility.** Dedicated MoM and YoY tracking for sales and profit surfaces momentum and seasonality at a glance.
4. **Granular drill-down.** Users can move from a company-wide summary to brand, product, customer segment, store state and individual customer delivery times.
5. **Logistics monitoring.** A dedicated page tracks online delivery lead time by month, customer and product.

12. Recommendations

1. Investigate the post-2019 decline in sales and profit. Margins held steady while orders fell by about half in 2020, so the drivers are likely volume related. Confirm first that 2020 and 2021 order data is complete.
2. Fix the data gaps. Obtain or reconstruct the orders missing between mid-March and late April each year before drawing conclusions about seasonality or month-over-month trends.
3. Prioritise profitable products and brands by looking at revenue, profit and margin together, not sales volume alone. Adventure Works, Contoso and Wide World Importers bring in 57% of revenue, while Tailspin Toys has the weakest margin (54.6%).
4. Grow the online channel. Its share of sales rose from 16.8% to 22.3% between 2016 and 2020 at the same margin as physical stores. Keep reducing lead time (4.5 days on average) and review slow items such as the Contoso Reserve Pen (15 days).

5. Strengthen high-performing markets and stores while investigating lower-performing locations to improve overall profitability.
6. Capture order and delivery locations (and, ideally, the dispatch date and the fulfilment point or courier) for every online order. At present, delivery lead time can only be compared by month, customer and product. With locations recorded, it can be broken down by region, route and fulfilment point, so the causes of slow deliveries, such as the 15-day lead time on the Contoso Reserve Pen, can be properly investigated. Recording the dispatch date would also split lead time into processing time and transit time.
7. Add inventory data: a stock table by product and store (product, store, date, quantity on hand, quantity received and stock-out date), related to both the Products and Stores tables. At present the model links products to stores only through sales, so it shows what was sold and where, but not what was in stock, when a product became available in a store, when it left, or when it should be reordered. With inventory data the report could track stock levels by store, flag stock-outs and low stock, and calculate reorder points so that new orders are placed before products run out.

13. Conclusion

This analysis reviewed sales performance across revenue, profitability, products, brands, customers, sales channels, store locations and delivery. Sales and profit rose through 2019 and then fell sharply in 2020 while margins stayed steady near 58.6%, which makes the post-2019 downturn the first issue to investigate.

Adventure Works is the largest brand by revenue, yet the best-selling individual product is a Wide World Importers desktop. Profit is heavily concentrated in the United States (53.55% by customer country), the online channel is growing at the same margin as physical stores, and online deliveries now take about 4.5 days. Differences in margin between brands show that sales volume alone is not enough to judge performance.

Combining revenue, profitability, product, customer, channel, store-level and delivery views gives a fuller picture, and the Power BI dashboard provides an interactive way to monitor it. Further analysis of the missing order periods, transaction-level location data, inventory data, customer behaviour and the drivers of the post-2019 decline would support deeper, more confident strategic decisions.

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